

Worksheet Problems on Data Structures and Algorithms

Worksheet 6

Q1: Count Inversions

Given an array of integers. Find the Inversion Count in the array. Two elements $arr[i]$ and $arr[j]$ form an inversion if $arr[i] > arr[j]$ and $i < j$.

Inversion Count: For an array, inversion count indicates how far (or close) the array is from being sorted.

If the array is already sorted then the inversion count is 0.

If an array is sorted in the reverse order then the inversion count is the maximum.

Examples:

Input: $arr[] = [2, 4, 1, 3, 5]$

Output: 3

Explanation: The sequence 2, 4, 1, 3, 5 has three inversions (2, 1), (4, 1), (4, 3).

Input: $arr[] = [2, 3, 4, 5, 6]$

Output: 0

Explanation: As the sequence is already sorted so there is no inversion count.

Input: $arr[] = [10, 10, 10]$

Output: 0

Explanation: As all the elements of array are same, so there is no inversion count.

Q2: Sort 0s, 1s and 2s

Given an array arr containing only 0s, 1s, and 2s. Sort the array in ascending order.

Examples:

Input: $arr[] = [0, 1, 2, 0, 1, 2]$

Output: $[0, 0, 1, 1, 2, 2]$

Explanation: 0s 1s and 2s are segregated into ascending order.

Input: $arr[] = [0, 1, 1, 0, 1, 2, 1, 2, 0, 0, 0, 1]$

Output: $[0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2]$

Q3: Minimum Platforms

Given arrival and departure times of all trains that reach a railway station. Find the minimum number of platforms required for the railway station so that no train is kept waiting.

Consider that all the trains arrive on the same day and leave on the same day. Arrival and departure time can never be the same for a train but we can have arrival time of one train equal to departure time of the other. At any given instance of time, same platform can not be used for both departure of a train and arrival of another train. In such cases, we need different platforms.

Note: Time intervals are in the 24-hour format(HHMM) , where the first two characters represent hour (between 00 to 23) and the last two characters represent minutes (this will be ≤ 59 and ≥ 0).

Input: arr[] = [0900, 0940, 0950, 1100, 1500, 1800],

dep[] = [0910, 1200, 1120, 1130, 1900, 2000]

Output: 3

Explanation: There are three trains during the time 0940 to 1200. So we need minimum 3 platforms.

Q4: Trapping Rain Water

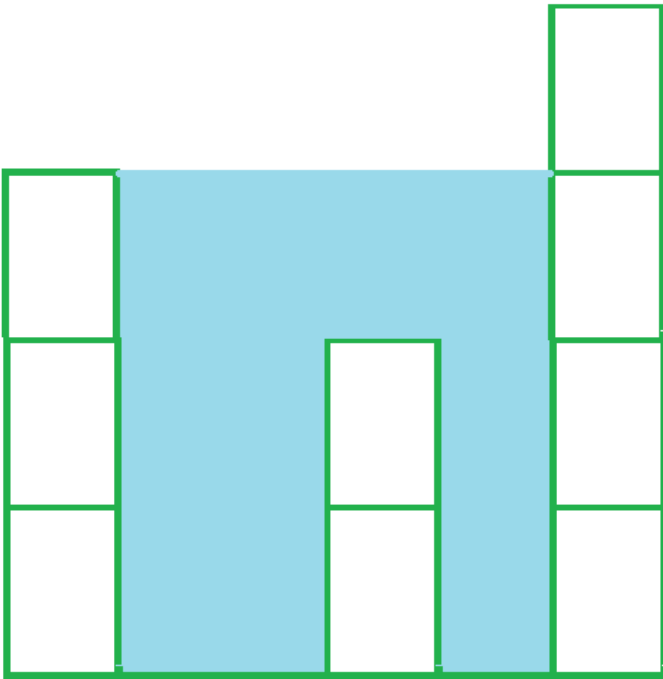
Given an array arr[] with non-negative integers representing the height of blocks. If width of each block is 1, compute how much water can be trapped between the blocks during the rainy season.

Examples:

Input: arr[] = [3, 0, 0, 2, 0, 4]

Output: 10

Explanation:



Bars for input {3, 0, 0, 2, 0, 4}

Total trapped water = 3 + 3 + 1 + 3 = 10

Input: arr[] = [7, 4, 0, 9]

Output: 10

Explanation:

Water trapped by above

block of height 4 is 3 units and above

block of height 0 is 7 units. So, the

total unit of water trapped is 10 units.

Input: arr[] = [6, 9, 9]

Output: 0

Explanation:

No water will be trapped.

Q5: Chocolate Distribution Problem

Given an array arr[] of positive integers, where each value represents the number of chocolates in a packet. Each packet can have a variable number of chocolates. There are m students, the task is to distribute chocolate packets among m students such that -

- i. Each student gets exactly one packet.
- ii. The difference between maximum number of chocolates given to a student and minimum number of chocolates given to a student is minimum and return that minimum possible difference.

Examples:

Input: arr = [3, 4, 1, 9, 56, 7, 9, 12], m = 5

Output: 6

Explanation: The minimum difference between maximum chocolates and minimum chocolates is $9 - 3 = 6$ by choosing following m packets :[3, 4, 9, 7, 9].

Input: arr = [7, 3, 2, 4, 9, 12, 56], m = 3

Output: 2

Explanation: The minimum difference between maximum chocolates and minimum chocolates is $4 - 2 = 2$ by choosing following m packets :[3, 2, 4].